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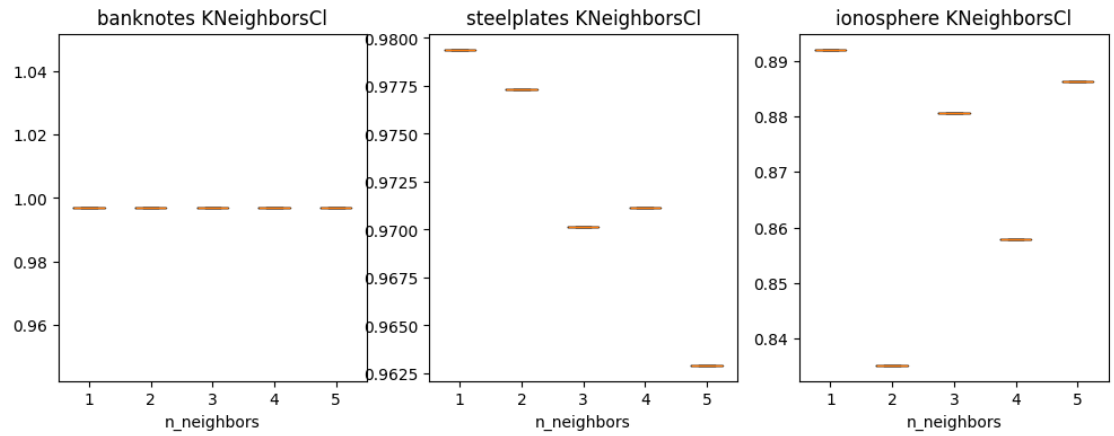
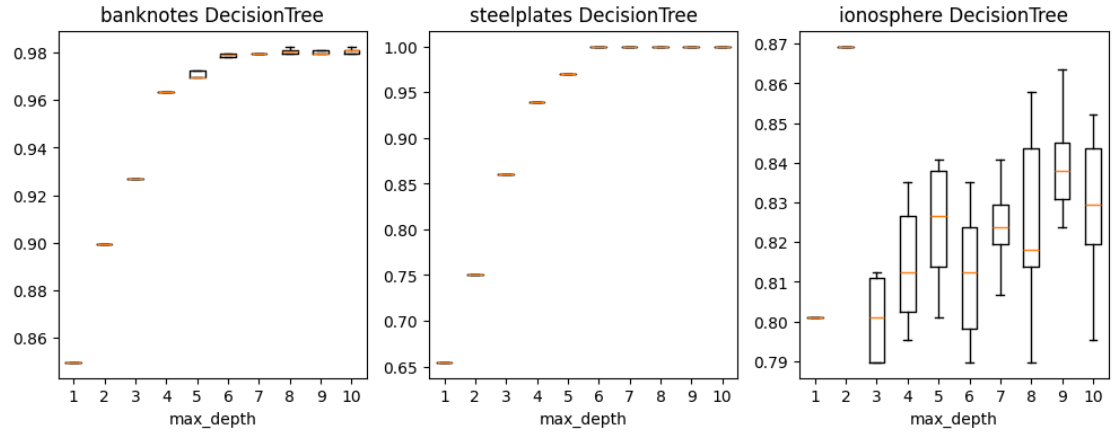
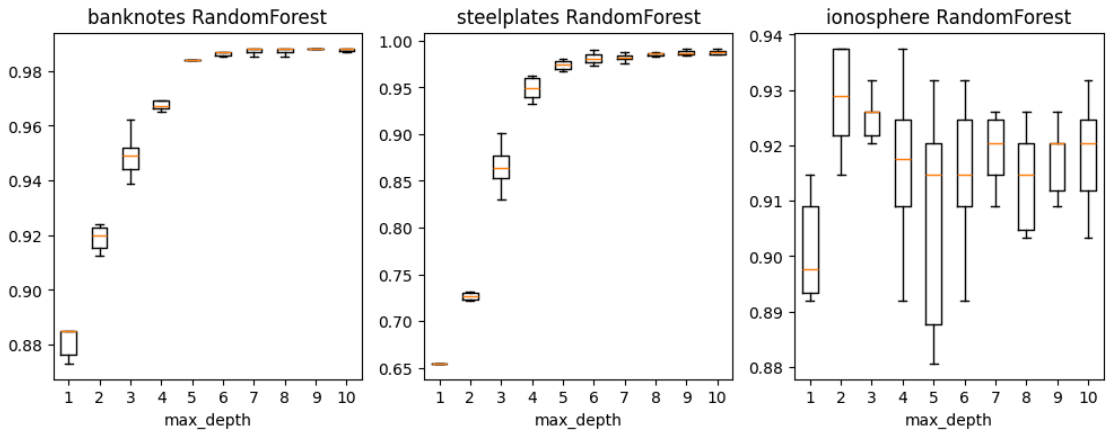
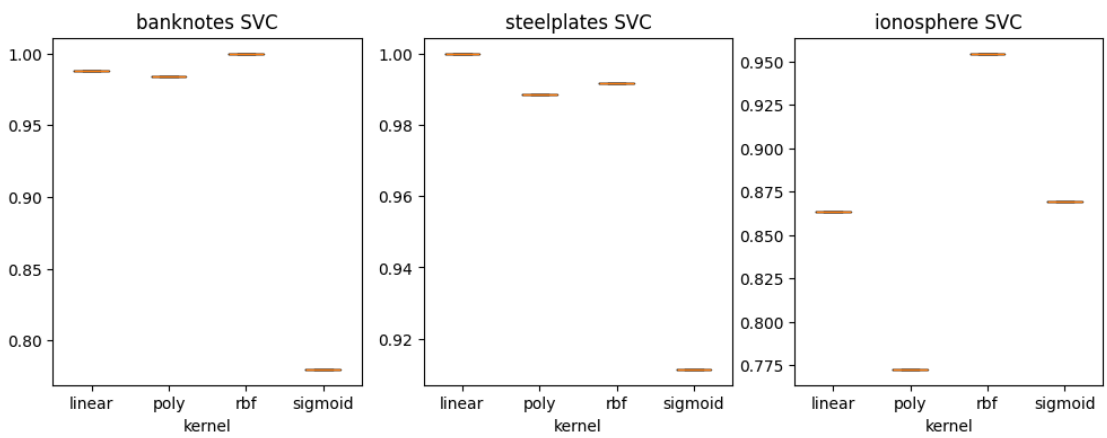
2026 AIML231 Techniques in ML: Assignment 1

Classification using the SKLearn library

Classification Accuracy versus Parameter Values

KNN	
	1.00 was the best KNeighborsCl on banknotes 0.98 was the best KNeighborsCl on steelplates 0.89 was the best KNeighborsCl on ionosphere Average = 0.96, performance range 0.11
DTC	
	0.98 was the best DecisionTree on banknotes 1.00 was the best DecisionTree on steelplates 0.87 was the best DecisionTree on ionosphere Average = 0.95, performance range 0.13

RFC	
	0.99 was the best RandomForest on banknotes 0.99 was the best RandomForest on steelplates 0.93 was the best RandomForest on ionosphere Average = 0.97, performance range 0.06
SVC	
	1.00 was the best SVC on banknotes 1.00 was the best SVC on steelplates 0.95 was the best SVC on ionosphere Average = 0.98, performance range 0.05

Best performance?

In terms of best performance, SVC scores perfect accuracy on two datasets, banknotes and steel plates, while still hitting high accuracy on ionosphere. All classifiers were able to make highly accurate predictions for the banknotes and steelplates datasets, but sharply dropped off on accuracy for predictions on the ionosphere dataset. From this we can also infer that the banknotes and steelplates datasets have a clean separation between clusters, while the ionosphere dataset does not, and has more noise than the other datasets.

Consistency?

In terms of most consistent performance, by comparing the range of accuracy for each classifier across the three datasets, SVC also takes first place, with the tightest performance range of 0.05. We can see from the charts that SVC also has little to no variance in accuracy scores in the 10 trials for each hyperparameter.

b) Discuss how sensitive each classifier's performance is to the changes in the hyperparameters and explain why based on model and dataset characteristics.

KNN
Range: 0.11
Sensitivity: moderate-highly sensitivity

K Nearest Neighbours has moderately high sensitivity across all three datasets with a range of 0.11 between highest accuracy scores. The range of accuracy scores between changes in hyperparameters for each dataset are:
banknotes: ~0
The low sensitivity to changes in n_neighbours with the banknotes dataset shows that the classes are cleanly separated - the increasing k with constant accuracy shows that each class is grouped closely together with the nearest neighbours almost always likely to share the same class.
steelplates: ~0.02
As k increases the accuracy decreases, and the sensitivity for KNN on steelplates slightly increases, which would indicate that the clusters here are not as cleanly separated as the banknotes clusters. KNN also starts to fall victim to the curse of dimensionality here, as the number of dimensions in the steelplates is 8 times greater than banknotes.
ionosphere: ~0.05
This has the highest sensitivity of the 3 model fits between changes in hyperparameters. The data has the same number of dimensions as the steelplates dataset, but worse accuracy and higher sensitivity, suggesting a very noisy dataset with well integrated clusters.
The lower performance in the steelplates and ionosphere datasets at higher k values also suggest model underfitting.

DTC
Range: 0.13
Sensitivity: highly sensitivity

banknotes: sharp curve up to max depth 6 before flattening out, showing moderate sensitivity on smaller depth levels, and low sensitivity on greater depth levels. The cleanly separated clusters mean deeper trees don't provide an equal value tradeoff in accuracy once key splits are established.
steelplates: sharper curve up to max depth 6 before flattening, showing high sensitivity on smaller depth levels, low sensitivity on greater depth levels . Lower accuracy on shallower trees show data has more complex patterns, but the plateau shows the patterns are structured.

ionosphere: accuracy spike at max depth 2, then drops back down with a slow increase, with large variance, both highly sensitive and unstable. The amount of noise in the data results in overfitting.

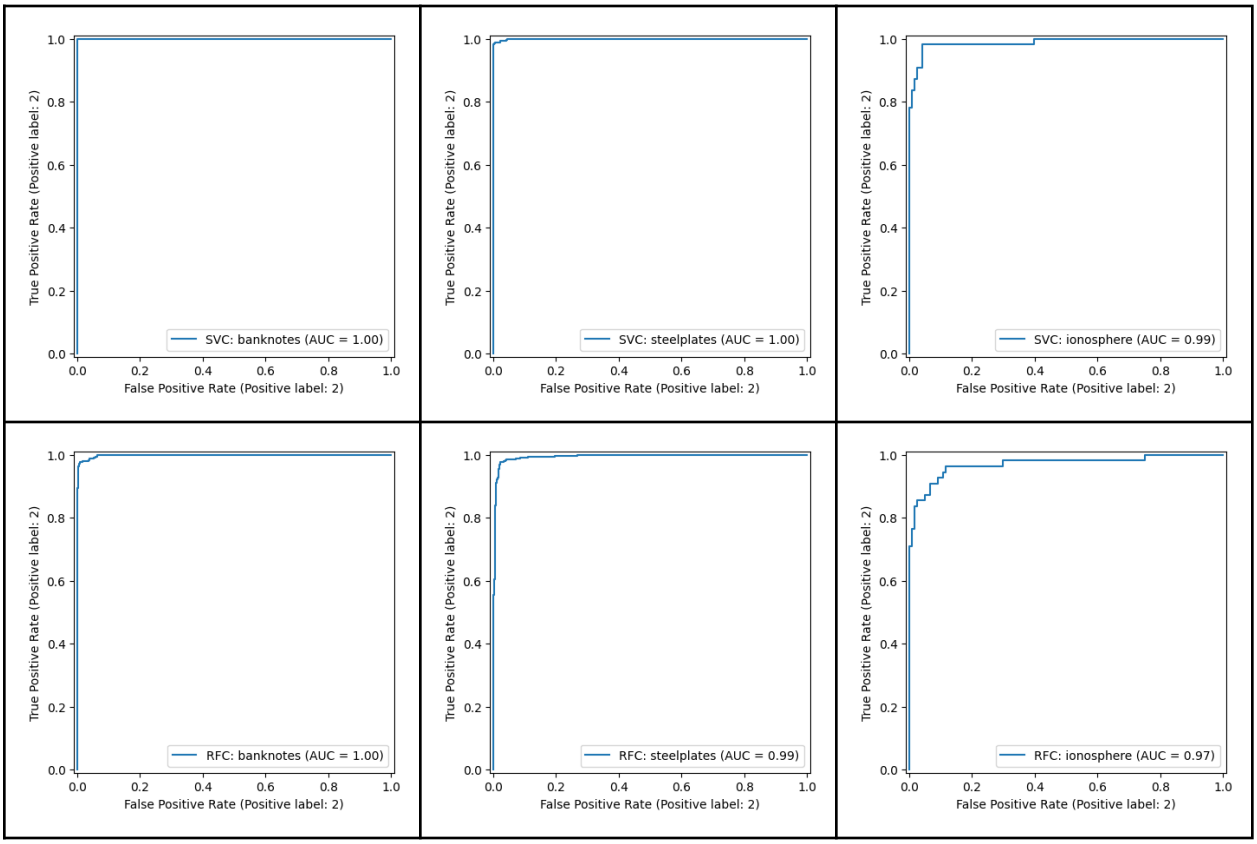
RFC
Range: 0.06
Sensitivity: low sensitivity

banknotes: sharp curve up to max depth 5 before flattening out, showing moderate sensitivity on smaller depths, and low sensitivity on greater depth levels. Again this is because the data is low dimensional and cleanly separated.
steelplates: steeper curve up to max depth 5 before flattening out, as above showing that the data is more complex with higher dimensions, but still contains useful patterns and clean separation.
ionosphere: highly sensitive. The great amount of noise in the data, the high dimensionality and the smaller number of observations result in overfitting and instability.

SVC
Range: 0.05
Sensitivity: low sensitivity

banknotes: Linear, polynomial, and RBF kernels all produce high accuracy scores, but sigmoid significantly underperforms. Sensitivity is low across meaningful kernels. High accuracy on linear suggests that the data is linearly separable, with no significant difference on accuracy from poly and RBF kernels supporting little non-linearity in the data. Sigmoid kernel behaves inconsistently if not carefully tuned, which is not required as other kernel selections suffice for the simpler dataset.
steelplates: Linear and RBF are highly accurate, polynomial less so, and sigmoid again significantly underperforms. Moderately complex dataset. There may be a greater degree of non-linearity in this dataset, and polynomial may under/overfit.
ionosphere: RBF scores highly, as it models localised decision boundaries. Incredibly complex data structure. Linear kernel underfits and polynomial may underfit as unable to appropriately capture structure and boundaries.

ROC Curve, for SVC and RFC by dataset



ROC Curve and AUC Comparison for Each Dataset

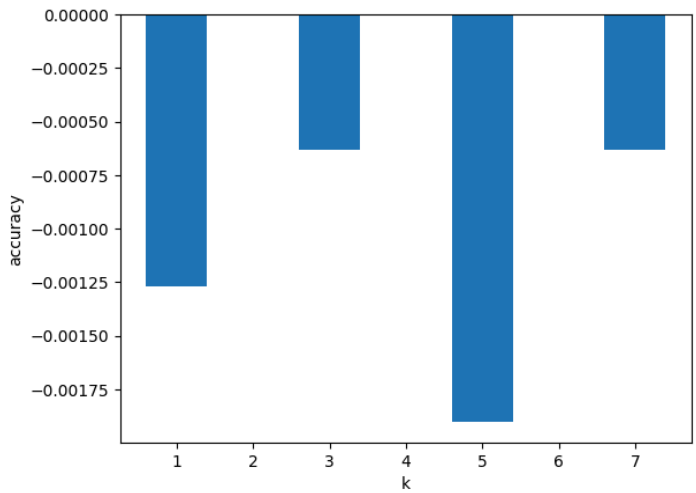
On banknotes both SVC and RFC show perfect accuracy, with ROC curves tucked tightly into the top left corner, and AUC values of 1.00. There is no meaningful difference between the classifiers on this dataset. On steelplates, the SVC curve has a sharper rise and stays closer to the top left compared to the RFC curve, and their AUC values are 1.00 and 0.99 respectively, showing SVC is a slightly better classifier than RFC for this data. On ionosphere, the SVC curve is sharper and hugs the corner much closer than the RFC curve, and the respective AUC values are 0.99 and 0.97. For this data, SVC performs significantly better than RandomForestClassifier.

K-Nearest Neighbor (k-NN) Classifier

Changing k and its effect on test accuracy

Finding the right value for k means searching for the balance point between overfitting and generalisation. From our test on banknotes, we see that k=1 is a fine accuracy, k=3 is the accuracy score, k=5 drops to the worst accuracy score, and k=7 returns to the same accuracy score as k=3.

For k=1, classifying on one neighbour can make the fit sensitive to noise. k=3 uses more neighbours, which helps the fit make more stable choices and can reduce the effect of noisy data points. k=5 broadens the data too much, and the decision boundary becomes smoother at the expense of precision in classification, and the neighbourhoods include more points from other classes. From this you would expect k=7 to show the same issue as k=5 but more exaggerated, but because the clusters in banknotes are clearly separated (as shown in part 1), adding more neighbours actually help to stabilise predictions.

The accuracy of the knn classifier is 0.9970845481049563 when k=1	
The accuracy of the knn classifier is 0.9985422740524781 when k=3	
The accuracy of the knn classifier is 0.9956268221574344 when k=5	
The accuracy of the knn classifier is 0.9985422740524781 when k=7	